

Semester/Year End Examination May-2023

Name of School: School of Engineering and Technology

Name of Programme/Course: B.Tech. (Biomedical) / B.Tech. (Biotech) / B.Tech. (Chemical) / B.Tech. (Civill) / B.Tech. (Mechanical) / B.Tech. (AIML) / B.Tech. (CSE-Cyber Security) / B.Tech. (Food Tech.) / B.Tec (Electrical)

Semester: II

Paper Code: BTEPHCO203T22

Title of Paper: Engineering Physics

Time: 3 hrs.

Max. M

INSTRUCTIONS/NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section-A and Section-B. S consists of 5 questions and Section-B consists of 6 questions. Attem questions from Section-A and any 4 questions from Section-B. Marks al mentioned against each question.

Question
Number

SECTION-A

1. What is interference of light? Describe Young's double slit experiment. (2)
2. Explain the phenomenon of polarization of light. (4)
3. Draw a neat and clean diagram of He-Ne laser and explain its construction. (4)
4. State and prove Heisenberg's uncertainty principle.
5. Explain the principle of optical fiber. (4)
6. Explain Einstein's Mass -Energy relation.

SECTION-B

1. Explain the Fraunhofer diffraction at single slit and derive an expression for intensity distribution due to Fraunhofer diffraction. — (6) 10
2. What are Einstein coefficients A and B? Derive relation between them. 10
3. Describe the step index and graded index fibers. Explain the difference between them. 10
4. Define Wave function and write its properties. Derive the time dependent Schrodinger wave equation. 10
5. Discuss Lorentz transformation. — (16) 10
6. Explain the gradient of a scalar field in detail. Discuss its physical significance. 10